Quadric Surfaces Practice Set

How to identify a quadric without memorising only the picture

Use this order:

simplify to standard form $\longrightarrow$ look for missing variables / sign pattern $\longrightarrow$ check two useful traces

Useful clues:

- a missing variable usually indicates a cylinder parallel to that axis;
- all squared terms with the same sign and equal to 1 suggest an ellipsoid;
- one unsquared variable with same-sign squares suggests an elliptic paraboloid;
- one unsquared variable with opposite-sign squares suggests a hyperbolic paraboloid;
- exactly one positive squared term in a hyperboloid equation gives a two-sheet hyperboloid, with its axis along that positive term;
- a homogeneous quadratic equation equal to 0 usually suggests a cone.

Problem 1

Classify the surface and find its intercepts

Identify the surface

$$4x^2 + 9y^2 + 36z^2 = 36.$$

Find its intercepts with the coordinate axes and describe its three coordinate-plane traces.

Detailed solution

First divide the equation by 36:

$$\frac{x^2}{9} + \frac{y^2}{4} + z^2 = 1.$$

All three variables are squared, all three terms have the same sign, and the right-hand side is 1. Therefore the surface is an **ellipsoid**.
Its semi-axis lengths are

$$3 \text{ along } x, \qquad 2 \text{ along } y, \qquad 1 \text{ along } z.$$

Hence the axis intercepts are

$$\boxed{(\pm 3, 0, 0), \quad (0, \pm 2, 0), \quad (0, 0, \pm 1)}.$$

For $z = 0$,

$$\frac{x^2}{9} + \frac{y^2}{4} = 1,$$

an ellipse in the xy-plane.
For $y = 0$,

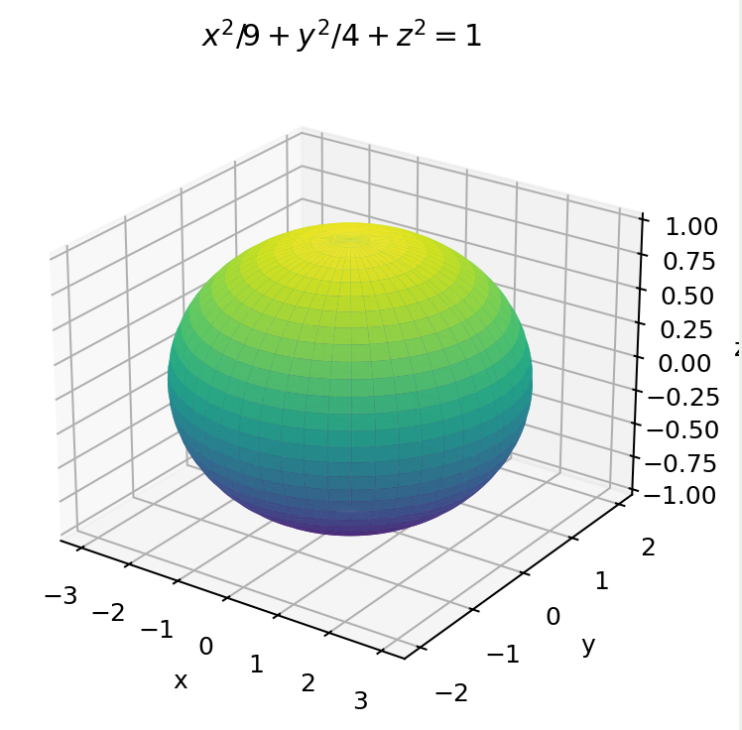
$$\frac{x^2}{9} + z^2 = 1,$$

an ellipse in the xz-plane.
For $x = 0$,

$$\frac{y^2}{4} + z^2 = 1,$$

an ellipse in the yz-plane.
Thus

$$\boxed{\text{the surface is an ellipsoid centred at the origin.}}$$



Quick check

A bounded quadric with three same-sign squared terms is a strong signal for an ellipsoid.

Problem 2

A cylinder that is not circular or elliptic

Identify and describe

$$z = (y-1)^2.$$

Which axis is the surface parallel to? Where is its "vertex" in three-dimensional space?

Detailed solution

The variable x does not appear anywhere in the equation.
Therefore, once a point (y, z) satisfies

$$z = (y-1)^2,$$

the coordinate x may take any real value.
In the yz-plane, the equation

$$z = (y-1)^2$$

is a parabola with vertex

$$(y, z) = (1, 0).$$

Because this same parabola is repeated for every value of x, the surface is a

$$\boxed{\text{parabolic cylinder parallel to the } x\text{-axis.}}$$

The "vertex" of a parabolic cylinder is not a single point. The vertex of the generating parabola sweeps out the line

$$\boxed{y = 1, \qquad z = 0, \qquad x \in \mathbb{R}.}$$

A useful trace check is

$$x = c \quad \Longrightarrow \quad z = (y-1)^2,$$

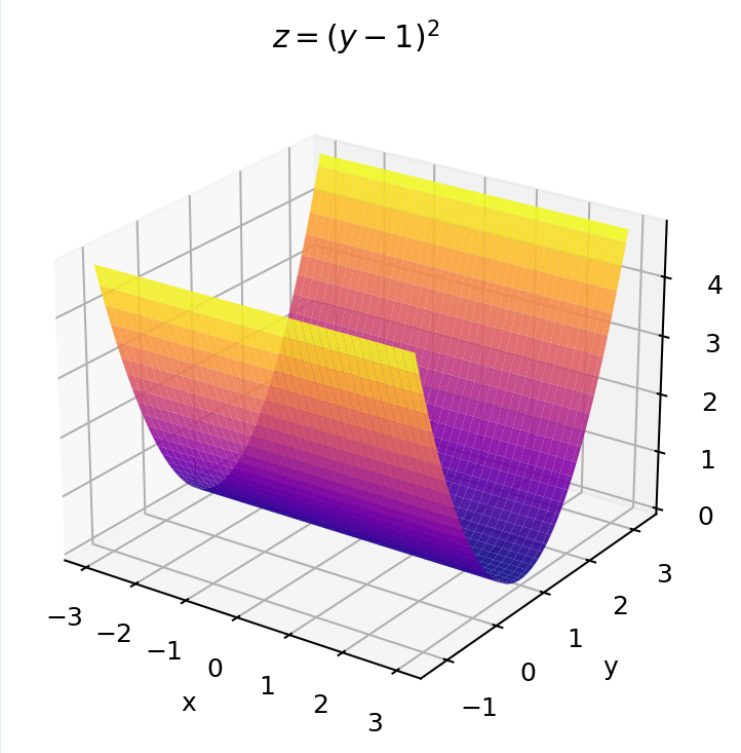
the same parabola for every constant c.
Also, if $z = k > 0$,

$$(y-1)^2 = k,$$

so

$$y = 1 \pm \sqrt{k}.$$

Thus a horizontal plane $z = k$ cuts the cylinder in two lines parallel to the x-axis.



Common mistake

A "cylinder" in multivariable calculus does not have to be circular. The essential clue is that one variable is missing, so a plane curve is copied parallel to the missing axis.

Problem 3

Identify a shifted quadric by completing squares

Identify the surface

$$x^2 + 4y^2 + z^2 - 4x + 8y - 6z + 8 = 0.$$

Find its centre and semi-axis lengths.

Detailed solution

Group the terms by variable:

$$(x^2 - 4x) + 4(y^2 + 2y) + (z^2 - 6z) + 8 = 0.$$

Complete the square in each variable:

$$x^2 - 4x = (x-2)^2 - 4,$$

$$4(y^2 + 2y) = 4\big((y+1)^2 - 1\big) = 4(y+1)^2 - 4,$$

and

$$z^2 - 6z = (z-3)^2 - 9.$$

Substitute:

$$(x-2)^2 - 4 + 4(y+1)^2 - 4 + (z-3)^2 - 9 + 8 = 0.$$

Therefore

$$(x-2)^2 + 4(y+1)^2 + (z-3)^2 = 9.$$

Divide by 9:

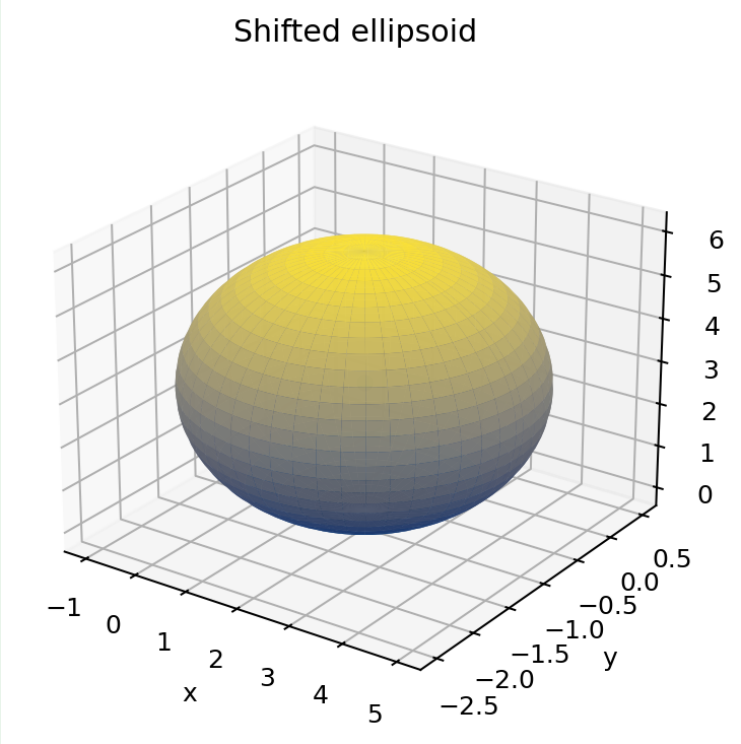
$$\frac{(x-2)^2}{9} + \frac{(y+1)^2}{9/4} + \frac{(z-3)^2}{9} = 1.$$

This is an ellipsoid centred at

$$\boxed{(2, -1, 3)}.$$

Its semi-axis lengths are

$$\boxed{3 \text{ along } x, \qquad \frac{3}{2} \text{ along } y, \qquad 3 \text{ along } z.}$$



Quick check

When linear terms such as $-4x$ or $8y$ are present, complete the square before classifying the surface.

Problem 4

A two-sheet hyperboloid with a different axis

Identify

$$\frac{x^2}{4} - \frac{y^2}{9} - z^2 = 1.$$

Determine its axis, explain why it has two separate sheets, and describe the traces $z = 0$, $y = 0$, and $x = 3$.

Detailed solution

The equation contains one positive squared term and two negative squared terms:

$$\frac{x^2}{4} - \frac{y^2}{9} - z^2 = 1.$$

For a two-sheet hyperboloid, the axis lies along the variable whose squared term is positive. Therefore the axis is the

$$\boxed{x\text{-axis.}}$$

Rewrite:

$$\frac{x^2}{4} = 1 + \frac{y^2}{9} + z^2.$$

The right-hand side is at least 1, hence

$$\frac{x^2}{4} \geq 1 \quad \Longrightarrow \quad |x| \geq 2.$$

So no point on the surface can satisfy

$$-2 < x < 2.$$

The surface therefore splits into the two pieces

$$\boxed{x \geq 2 \qquad \text{and} \qquad x \leq -2.}$$

For $z = 0$,

$$\frac{x^2}{4} - \frac{y^2}{9} = 1,$$

a hyperbola in the xy-plane.
For $y = 0$,

$$\frac{x^2}{4} - z^2 = 1,$$

a hyperbola in the xz-plane.
For $x = 3$,

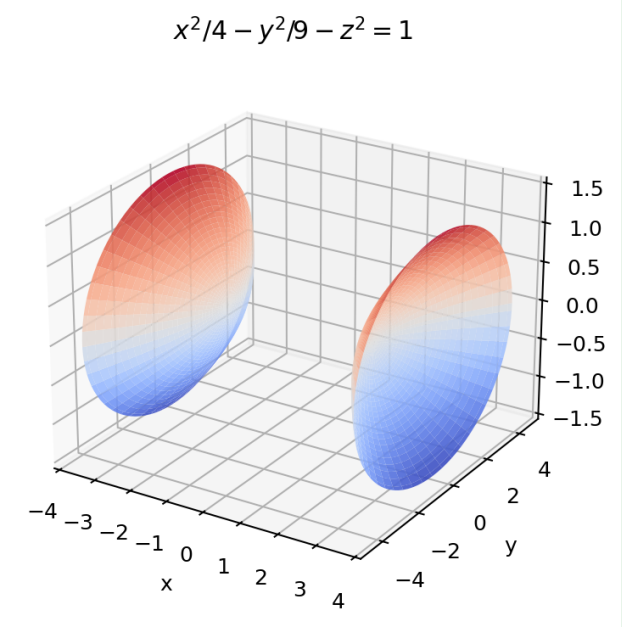
$$\frac{9}{4} - \frac{y^2}{9} - z^2 = 1,$$

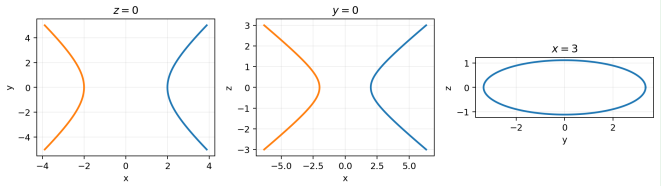
hence

$$\frac{y^2}{9} + z^2 = \frac{5}{4},$$

an ellipse in the plane $x = 3$.
Therefore

$$\boxed{\text{the surface is a two-sheet hyperboloid with axis along } x.}$$





Problem 5

Recognising the direction of a saddle

Identify

$$x = \frac{y^2}{4} - \frac{z^2}{9}.$$

Use traces to explain the geometry and determine the direction in which the saddle opens.

Detailed solution

The variable x is unsquared, while the two squared variables y^2 and z^2 have opposite signs. Therefore the surface is a

$$\boxed{\text{hyperbolic paraboloid.}}$$

Because x is the unsquared variable, the basic opening behaviour is along the x-direction.
Set $z = 0$:

$$x = \frac{y^2}{4},$$

a parabola opening toward positive x.
Set $y = 0$:

$$x = -\frac{z^2}{9},$$

a parabola opening toward negative x.
These two perpendicular parabolic traces open in opposite directions, producing the saddle.
For $x = 1$,

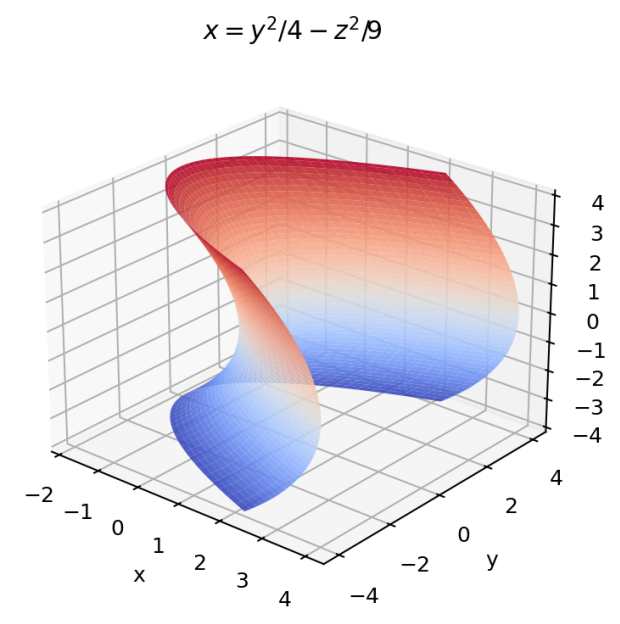
$$1 = \frac{y^2}{4} - \frac{z^2}{9},$$

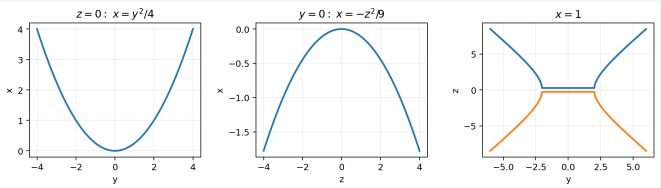
so

$$\frac{y^2}{4} - \frac{z^2}{9} = 1,$$

a hyperbola.
Thus

$$\boxed{\text{the surface is a hyperbolic paraboloid organised along the } x\text{-direction.}}$$





Quick check

For a hyperbolic paraboloid, two perpendicular parabolic traces opening in opposite directions are an excellent confirmation.

Problem 6

Build the equation from geometric information

A double cone has vertex at the origin and axis along the z-axis. Its intersection with the plane $z = 6$ is

$$\frac{x^2}{9} + \frac{y^2}{4} = 1.$$

Find an equation of the cone. Then describe its cross-section at $z = 3$.

Detailed solution

A cone with vertex at the origin has similar cross-sections at different heights. Therefore the semi-axis lengths scale directly with $|z|$.
At $z = 6$, the cross-section has semi-axis lengths

$$3 \text{ in the } x\text{-direction}, \qquad 2 \text{ in the } y\text{-direction}.$$

Hence at a general height z, the corresponding semi-axis lengths are

$$\frac{|z|}{2} \quad \text{and} \quad \frac{|z|}{3},$$

because

$$\frac{6}{2} = 3, \qquad \frac{6}{3} = 2.$$

Thus a general cross-section satisfies

$$\frac{x^2}{z^2/4} + \frac{y^2}{z^2/9} = 1.$$

Multiply through by z^2:

$$4x^2 + 9y^2 = z^2.$$

An equivalent standard form is

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = \frac{z^2}{36}.}$$

Check at $z = 6$:

$$\frac{x^2}{9} + \frac{y^2}{4} = 1,$$

as required.
Now put $z = 3$:

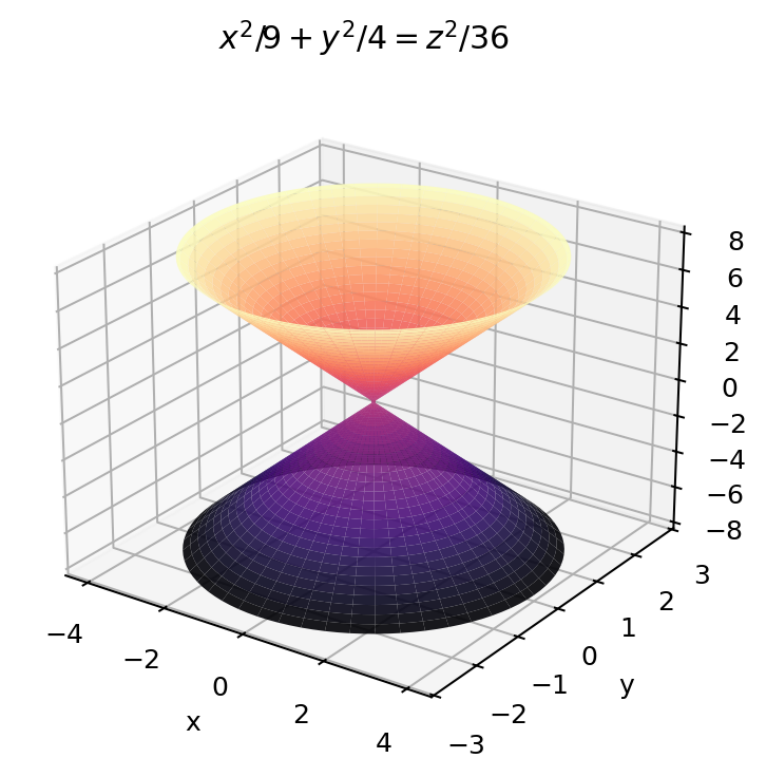
$$\frac{x^2}{9} + \frac{y^2}{4} = \frac{1}{4}.$$

Equivalently,

$$\frac{x^2}{9/4} + \frac{y^2}{1} = 1.$$

Therefore the cross-section at $z = 3$ is an ellipse with semi-axis lengths

$$\boxed{\frac{3}{2} \text{ and } 1.}$$



Quick check

This reverses the usual task: instead of identifying a surface from its equation, we construct the equation from its geometric cross-sections.

Final quick comparison

Surface	Main clue	Useful confirming trace
Ellipsoid	three same-sign squared terms; bounded	coordinate-plane ellipses
Parabolic cylinder	one variable missing; remaining equation is parabolic	fixing the missing variable gives the same parabola
Two-sheet hyperboloid	exactly one positive square in standard form	perpendicular traces are hyperbolas; sections beyond the gap are ellipses
Hyperbolic paraboloid	one unsquared variable; squared terms have opposite signs	perpendicular parabolic traces open in opposite directions
Elliptic cone	homogeneous quadratic relation	sections perpendicular to the axis are similar ellipses